

assigned an IP address.

Say you've specified the IP address of a Web site you wish to access. A server usually has a static IP address. However, your home PC, if you're connecting to the Internet through a modem, is usually assigned a different IP address by the ISP each time you dial in. This IP address is unique to you as a user, but only for as long as the session lasts. The reason for this system is that ISPs have to deal with a large number of requests at the same time. Besides, at any given time, the number of users of the ISP server is limited. To assign permanent IP addresses to retail users wouldn't really be required; instead, ISPs simply allot any IP address that is not being used at the time to each user who has just dialled in. This system enables the ISP to get along with fewer IP addresses for the number of users who are connected simultaneously.

TCP breaks down and reassembles packets of data. IP ensures that the packets are sent to the right destination. The TCP/IP combination is used because the Internet is a "packet-switched network." In a "circuit-switched network," in contrast, once a connection is made, that specific part of the network is exclusively used only for that connection. In a packet-switched network, the connection between the sender and the receiver is not single and unbroken. When information is sent, it is broken into small packets and sent over many different routes at the same time and then put together in order at the other end once the packets of data reach the destination.

2.1.9 How TCP/IP works

Data sent across the Internet is broken up into packets of less than about 1,500 characters each. Each packet is given a "header," which holds the information needed to put the packets back in right order again. To each header is added a "checksum"—basically a number that checks whether the precise amount of data in the packet has been received or not. Each packet is put into what is called an IP "envelope"—a packet of data with a common address and other specifications. Once the packets go through the routers to their destination, TCP, at the receiver's end, calculates a checksum for each packet and then compares it with the checksum that